

Creative Infrastructure as Ideology: A Feminist Critique of Designing Creativity Support Tools In The Age of Generative AI

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1 Introduction

In 2022, a visual artwork titled ‘Théâtre D’opéra Spatial’ bagged the top prize at Colorado State Fair’s fine art competition however spurred controversy as many have objected over the use of Generative AI (GenAI) in art.¹ In this age that everyone is talking about Artificial Intelligence (AI), practitioners in creative industries emerge as front runners of using and being affected by GenAI (Sun et al., 2024). As they benefit from the escalating efficiency, creativity, and productivity these tools provide with, controversies and conflicts inevitably follow, from ethics discussion on automation of creative work (Vartiainen & Tedre, 2023), anxiety of job displacement within creative economics (Erickson, 2024) to legal disputes around copyrightability (Kyi et al., 2025) and compensation (Seger et al., 2023).

Among other practitioners, it has been a convention that artists construct and assert selfhood on their own terms, counter marginalised situations and claim authorship over their own representation. AI-integrated Creativity Support Tools (CSTs) enter this field not as a neutral instrument, but carries values from its makers and restructuring the conditions under which creative identity is built. When an artist's style becomes training data, they are already inside a relationship before they decide whether to enter or not. Yet the growing body of Human-Computer Interaction (HCI) research on CSTs largely frames this as an adoption problem, persistently asking how to design tools artists will use and even engineer adoption through user psychology, however being silent about the underlying ideologies those tools carry and how this may harm what artists really value.²

This essay argues that AI-integrated CSTs are not neutral instruments but carriers of specific ideologies and conflicts with how visual artists construct and maintain their creative identity. Drawing on technofeminism and broader feminist science and technology studies (STS),

¹ *Théâtre D’opéra Spatial*, by Jason M. Allen using Midjourney, won the 2022 Colorado State Fair’s Digital Arts/Digitally-Manipulated Photography category for emerging artists.
<https://www.firstpost.com/explainers/explained-the-controversy-surrounding-the-ai-generated-artwork-that-won-us-competition-11188431.html>

² The discussion in this essay is around visual artists in specific.

Section 2 introduces the theoretical framework for examining the gendered infrastructure of creativity. Section 3 interrogates specific ideologies embedded in AI-integrated CST design and traces their implications for artists' identity and labour. Section 4 reads and interprets values reflected in artists' responses as design evidence, whereas Section 5 argues that feminist design methods offer an alternative framework for CST design that centres artists' values rather than overriding them.

2 The Gendered Infrastructure of Creativity

Among creativity-oriented HCI studies, a growing body of literature focuses on the design and evaluation of computational techniques that support users across various creative phases (Frich et al., 2019). As the infrastructure that supports creative practice from ideation, exploration, execution to dissemination, scholars have developed tools ranging from brainstorming aids, generative design systems to specific applications, and corresponding evaluation frameworks oriented around system usability, creative output and user adoption.

Feminist HCI scholars critique the design of these tools are not neutral (Hsueh et al., 2024), however, reflects specific epistemic assumptions about what creative work is, who the creative subject is and what role technology should play. These assumptions are produced by specific social actors and serve specific interests. For example, amid the technology market, a dominant position in commercial CST design treats creative work as *tool-mediated expert activity* which embodies a vision of technology that equates creativity with the creativity support tool itself and nudges practitioners' adoption.

To examine implicit ideologies and their consequences for artists' identity construction, we draw from Wajcman's (2010) social constructivist account of technology, in which technological systems are understood as materially encoding the social relations of their production, and ask *what values AI-integrated CST design embeds, whose interests those values serve, at whose expense, and what alternatives become visible when read through artists' confrontation.*

3 Questioning Ideologies Embedded in AI-Integrated CSTs

With the proliferation of GenAI, CST research has increasingly focused on integrating it into specific creative workflows. Tools like Midjourney, Adobe Firefly, and DALL-E are now

embedded in professional creative practice (Sun et al., 2024), and a corresponding body of HCI research has followed, with a focus on designing creative, mixed-initiative systems and exploring how AI can be a co-agency in between (Hu et al., 2025). What the integration of AI amplifies, however, is not only the capabilities of the tool but the assumptions already embedded in and the design choices carrying specific values.

In the technology market, the aforementioned vision of *tool-mediated expert activity view* revolves around discourses like “[tools] supercharge your creativity” “creative copilot” “[tools] unlocking creative potential”, and is heavily promoted by tech giants, from creativity-specialised industries like Adobe and Autodesk, to generalist industries like OpenAI and Google. This dominant voice implicitly suggests that the more fluently a professional uses the tool, the more creative they become. Similar emphasis is also evident in how CSTs are evaluated. Commonly used evaluation measures are established around system usability and creative artefact quality, however only a small fraction is about user-centric benefits, for example intrinsic abilities, emotional well-being, self-reflection and other self-perception measures (Rhys Cox et al., 2025). These reflect the underlying values that efficiency, productivity, and seamless adoption are self-evidently desirable outcomes, and that tools which deliver these outcomes are by definition good tools.

There is also a trend that conceptualises AI-integrated CSTs for the democratisation of creativity, which claims to lower the barriers to creative expression so that anyone can be artists. This seemingly ethically sound vision however conceals what Haraway (1988) criticised as god trick – the presentation of a situated, interested knowledge claim as a view from nowhere, omniscient and unbiased. The democratisation narrative is produced from the position of actors who benefit from maximum adoption, yet presents itself as a universal good, masking whose interests it serves and at whose expense. What it implicitly assumes is that creative identity is freely reproducible and available for extraction and redistribution. This assumed position is what makes style appropriation legible as democratisation for which we see evidence in growing lawsuits against GenAI companies³, protests from creative workers⁴ and the ongoing battle over intellectual property rights.

³ Written Testimony of Karla OrtizUS. Senate Judiciary Subcommittee on Intellectual Property“AI and Copyright” July 7th, 2023 https://www.judiciary.senate.gov/imo/media/doc/2023-07-12_pm_-_testimony_-_ortiz.pdf

⁴ Los Angeles Times, “Writers’ strike: What to know,” May 1, 2023.

<https://www.latimes.com/entertainment-arts/business/story/2023-05-01/writers-strike-what-to-know-wga-guild-hollywood-productions>

A directly visible benefit of “escalating efficiency” and “democratised creativity” promised by tech companies is that everyone can focus on “what matters most”. Is the assumption of AI-integrated CSTs liberating productivity justifiable? Cowan’s (1976) historical analysis empirically grounded that the development of household technology did not actually liberate women from domestic work and free up their time, however increased the workload and responsibility of middle-class housewives as well as the emotional stakes of their work. Similar evidence is also seen in the trajectory of GenAI into creative industries. Not only 95% of organisations that integrate the use of GenAI technologies in their processes see no measurable return on their investment in these (NANDA, 2025), but they often end up dealing with workslops (Niederhoffer, 2025) that AI-generated poor quality work requires time to be fixed and erodes trust among co-workers. Tools that promise to free creative workers from repetitive labour instead generate new forms of invisible labour, from back and forth prompt engineering, iterating through unpredictable outputs to manually correcting AI-generated content before use (Sun et al., 2024).

4 Artists' Responses as Design Evidence

Prior work documents how individuals experiment, set boundaries, selectively integrate GenAI as they attempt to reconcile its opportunities with its perceived threats, or appropriating its capabilities to reimagine alternative technological trajectories.

- Embrace the development of GenAI, integrating them in their creative workflow.
- Resistance through a variety of counter practices to claim their human art identity and agency.
- Negotiating the use of GenAI in their workplace (Jiang et al., 2026)

Analyse artist-centered values reflected in between.

5 Toward Feminist CST Design In the Age of Generative AI

Compare between values reflected in problematic ideologies in current CSTs design with those evident in artists’ responses.

Argue the design of CSTs should not simply prioritise the aforementioned values since they conflicted with artist group’s covert values. What they seem to want may not actually be good for them and can harm their long-term interest (Sharma et al., 2026).

Propose feminist approaches towards real user-centred CST design in the age of GenAI

- Value negotiation
- Participatory and feminist design methods
- New measures for creativity support
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6 Limitations

7 Conclusion

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